PAPER_1603 — Jupiter Mass / Earth Mass = M_J/M_⊕ = D_crit·SO_5 + SSQ·SO_5 + SO_5·D_phys + SO_5 + K_MEX = 260 + 5.7 + 40 + 10 + 2.083 = 317.78

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astronomy **Date:** June 18, 2026 **Status:** CLOSED — 0.005% integer-primitive identity

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Observation

PAPER_1209Z_S580 (Astronomy Unified Proof Set) lists **Jupiter Mass / Earth Mass** with the integer-primitive expression:

`M_J/M_⊕ = D_crit·SO_5 + SSQ·SO_5 + SO_5·D_phys + SO_5 + K_MEX = 260 + 5.7 + 40 + 10 + 2.083 = 317.78`

Residual: **0.005%**.

UQFF Closed Identity

`M_J/M_⊕ = D_crit·SO_5 + SSQ·SO_5 + SO_5·D_phys + SO_5 + K_MEX = 260 + 5.7 + 40 + 10 + 2.083 = 317.78`

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical astronomy measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209Z_S580
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "jupiter_mass_ratio_317_8"})`
